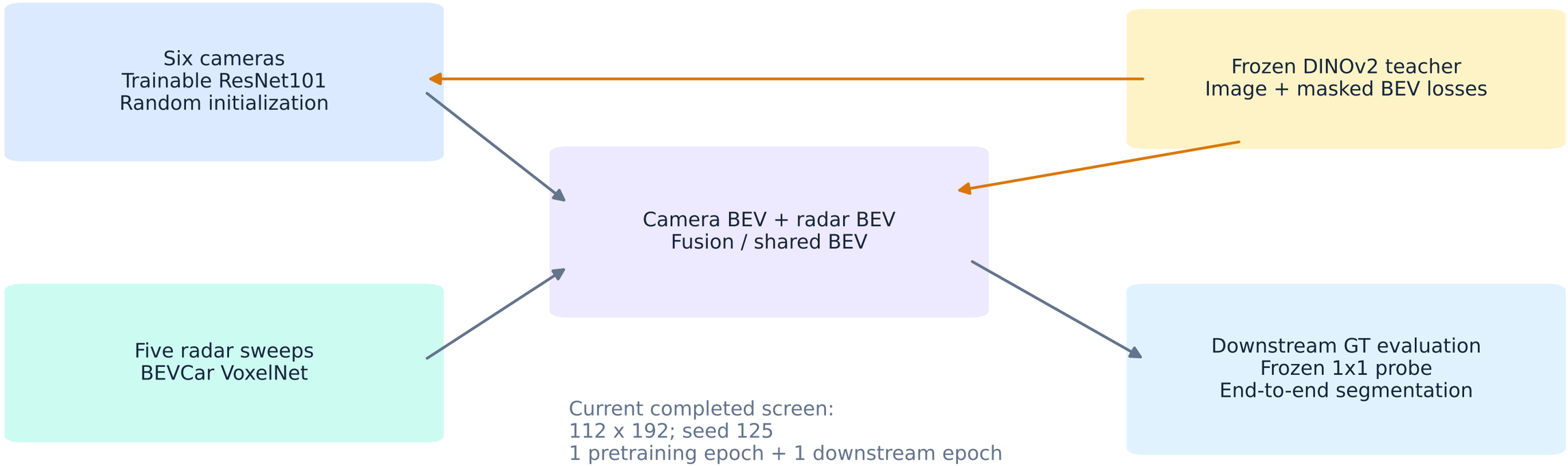
What we built and what we can measure

Full dataset: 28,130 training frames / 700 scenes; 6,019 validation frames / 150 scenes



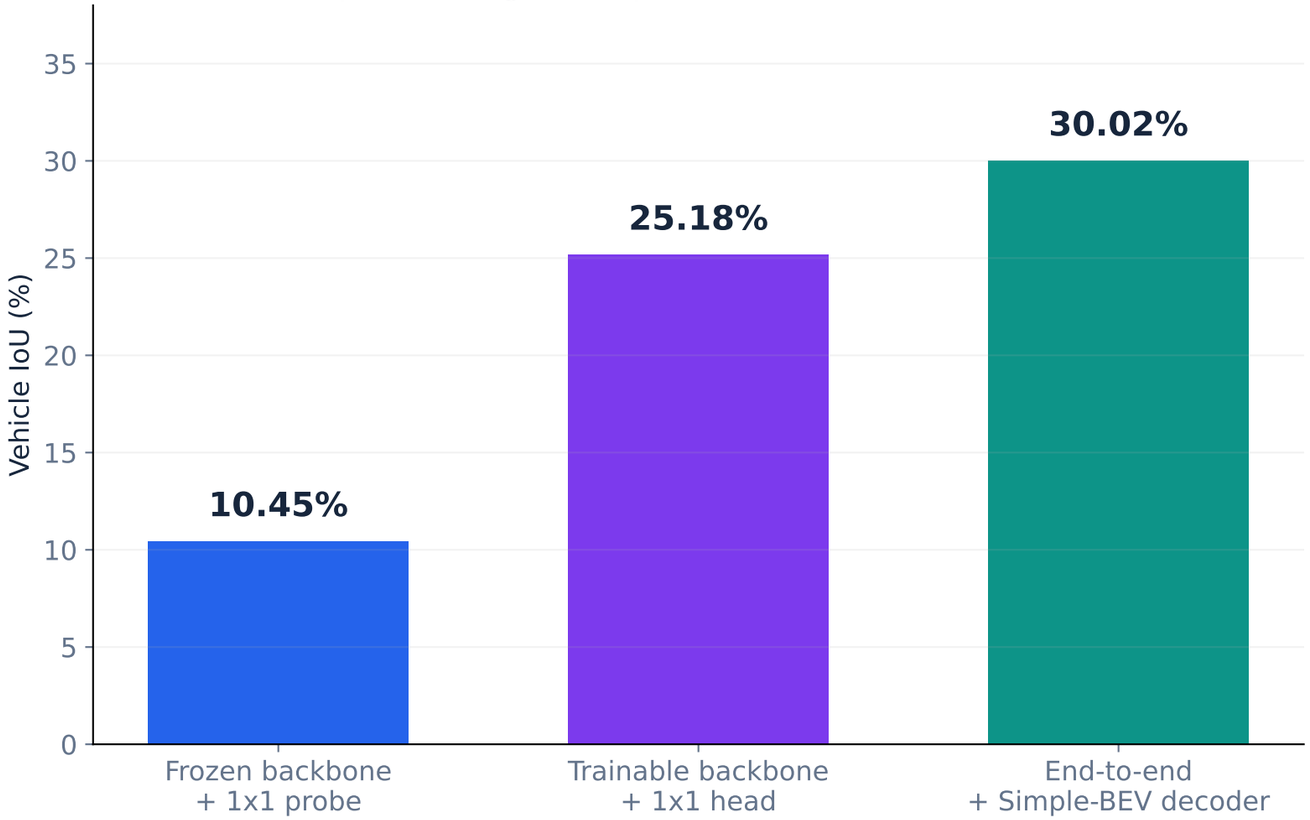
Current completed screen:
112 x 192; seed 125
1 pretraining epoch + 1 downstream epoch

Two complementary questions: is the representation readable, and how well can the system adapt?

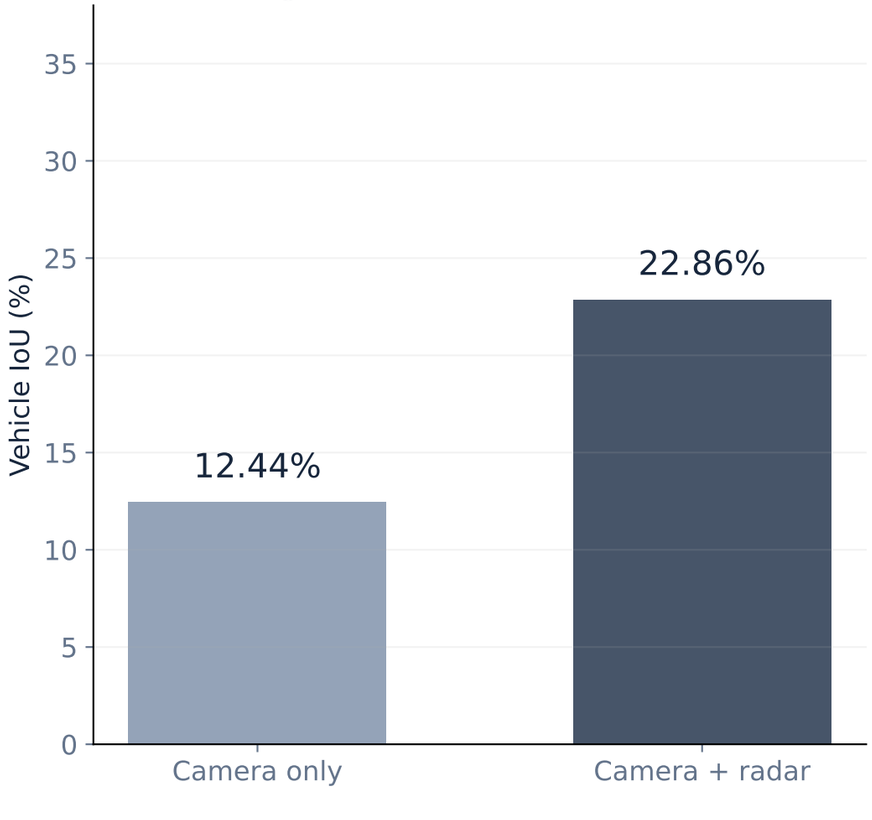
Vehicle segmentation: two evaluation protocols

Global valid-cell Vehicle IoU = TP / (TP + FP + FN); sigmoid threshold 0.5; all 6,019 validation frames

Same student pretraining checkpoint



Official weights at 112 x 192



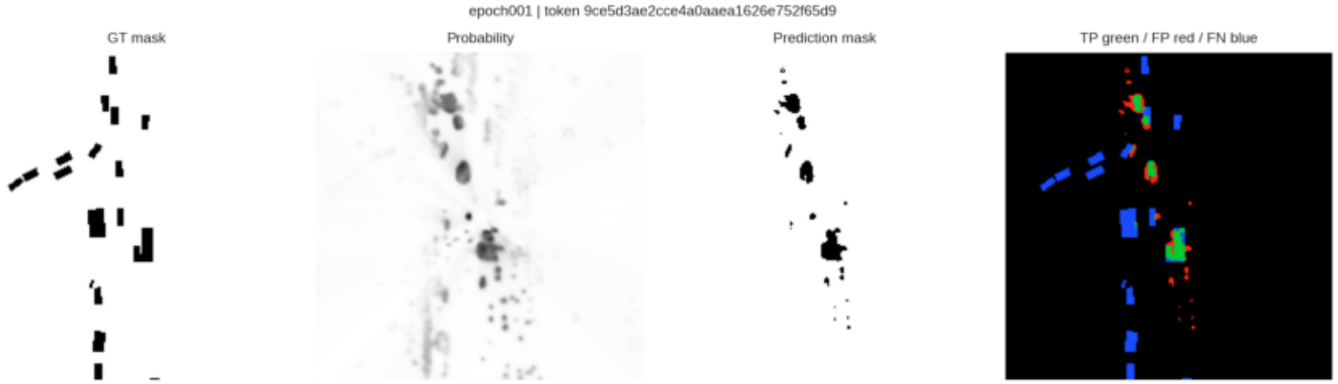
Official checkpoints were trained at higher resolution.
These are resolution-stress references, not matched retraining.

End-to-end adaptation improves IoU; this comparison does not isolate the benefit of DINO pretraining.

Prediction masks versus GT: fixed validation scene 1

Token 9ce5d3ae2cce4a0aaea1626e752f65d9 | Identical scene, labels, threshold and BEV grid for both readouts

Frozen backbone + trained 1x1 probe | frame IoU 21.07%



Trainable backbone + Simple-BEV decoder | frame IoU 44.78%

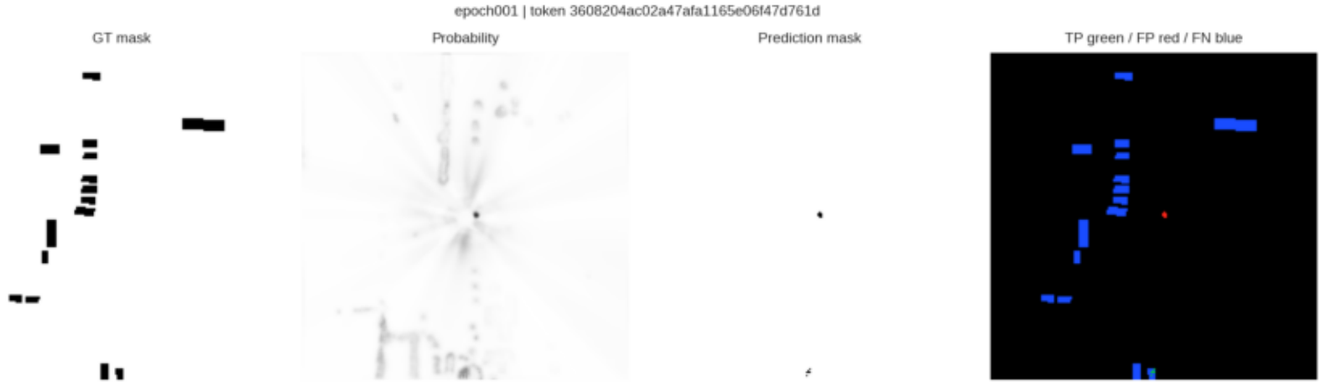


Error colors: green = correct vehicle; red = false positive; blue = missed vehicle. Black in mask = vehicle.

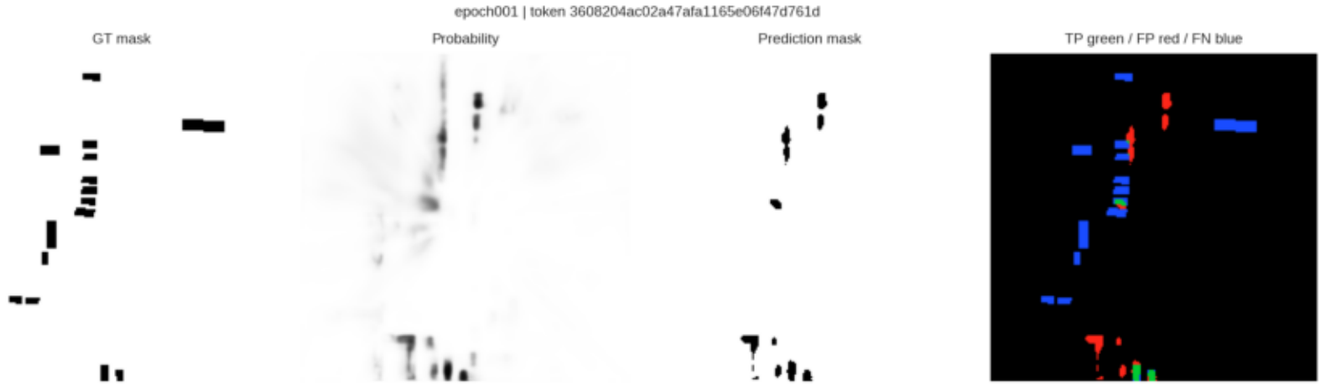
Prediction masks versus GT: fixed validation scene 2

Token 3608204ac02a47afa1165e06f47d761d | Identical scene, labels, threshold and BEV grid for both readouts

Frozen backbone + trained 1x1 probe | frame IoU 0.58%



Trainable backbone + Simple-BEV decoder | frame IoU 8.47%

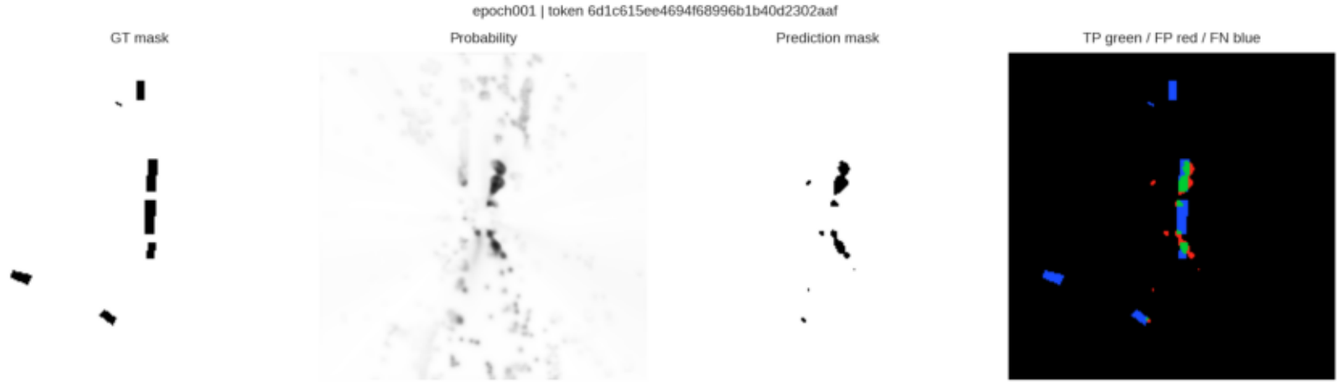


Error colors: green = correct vehicle; red = false positive; blue = missed vehicle. Black in mask = vehicle.

Prediction masks versus GT: fixed validation scene 3

Token 6d1c615ee4694f68996b1b40d2302aaf | Identical scene, labels, threshold and BEV grid for both readouts

Frozen backbone + trained 1x1 probe | frame IoU 23.23%



Trainable backbone + Simple-BEV decoder | frame IoU 38.13%

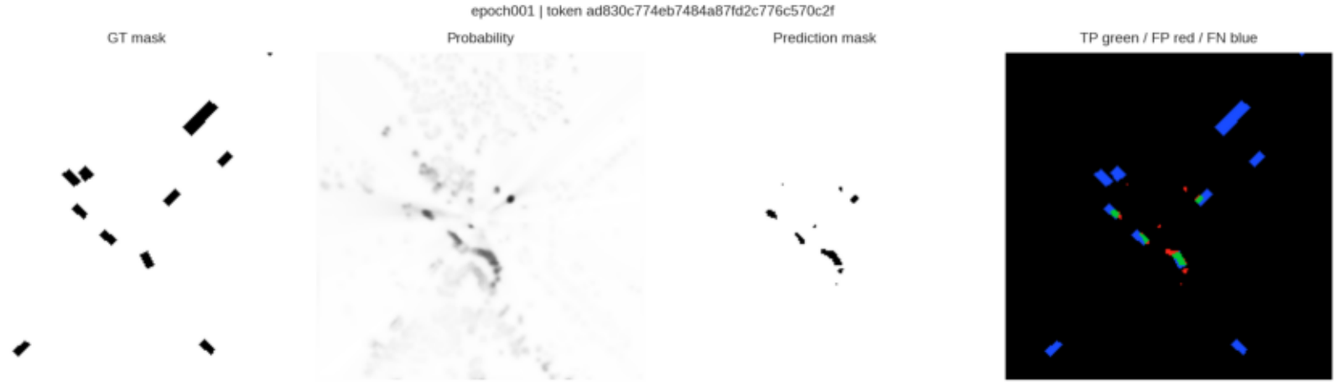


Error colors: green = correct vehicle; red = false positive; blue = missed vehicle. Black in mask = vehicle.

Prediction masks versus GT: fixed validation scene 4

Token ad830c774eb7484a87fd2c776c570c2f | Identical scene, labels, threshold and BEV grid for both readouts

Frozen backbone + trained 1x1 probe | frame IoU 13.03%



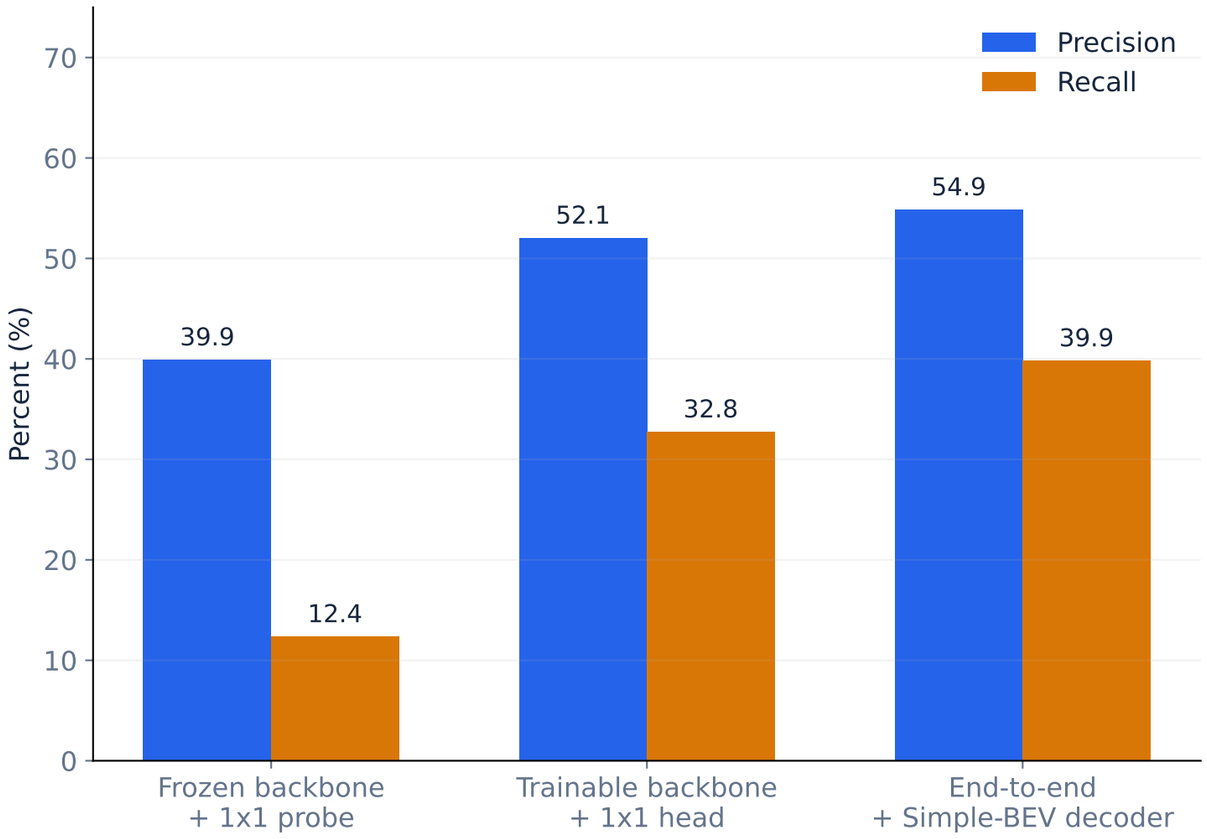
Trainable backbone + Simple-BEV decoder | frame IoU 23.50%

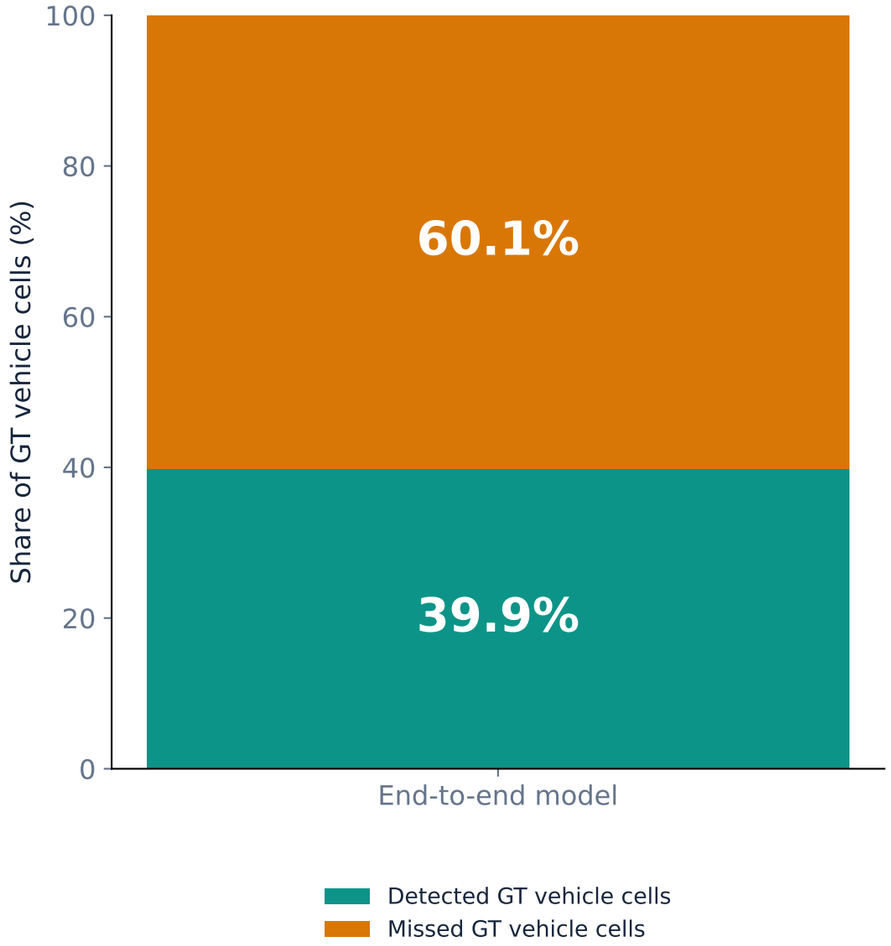


Error colors: green = correct vehicle; red = false positive; blue = missed vehicle. Black in mask = vehicle.

What limits segmentation right now?

Absolute confusion counts reveal errors that a single IoU number hides

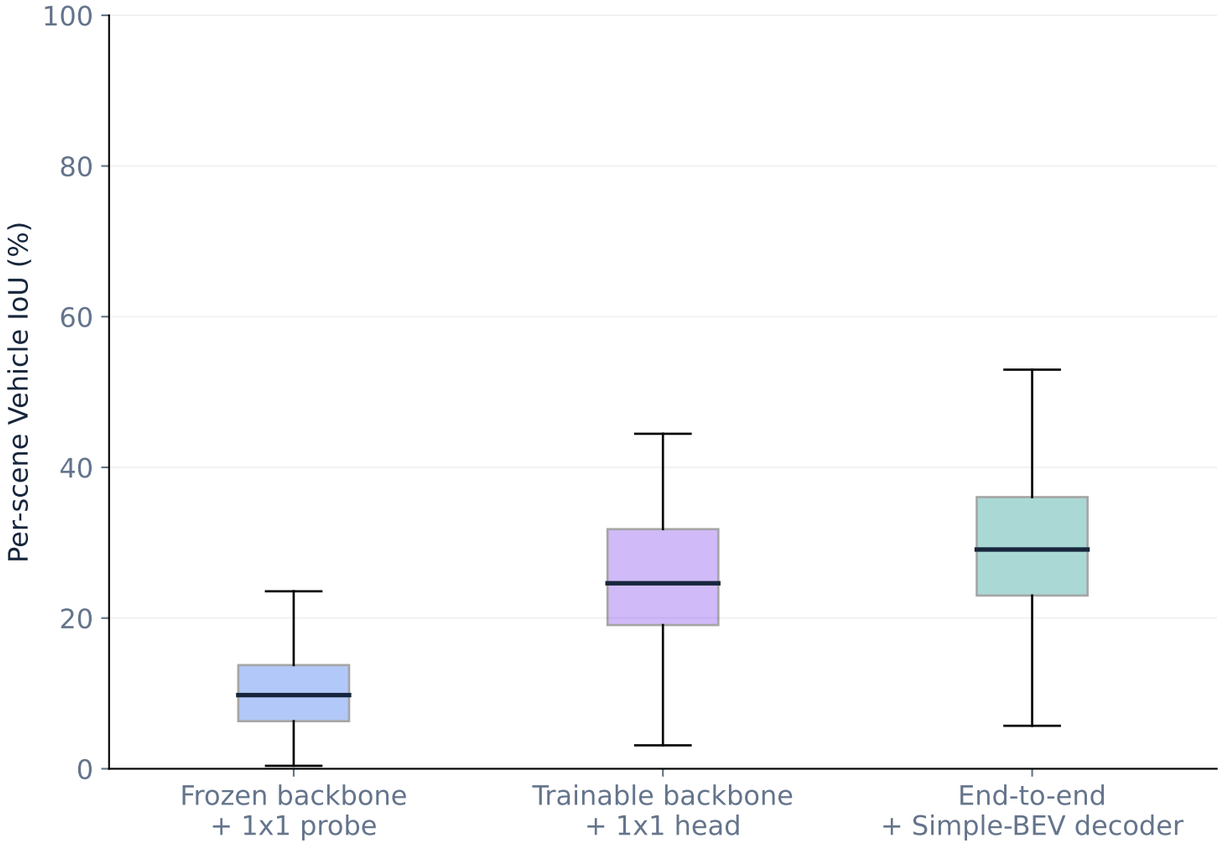


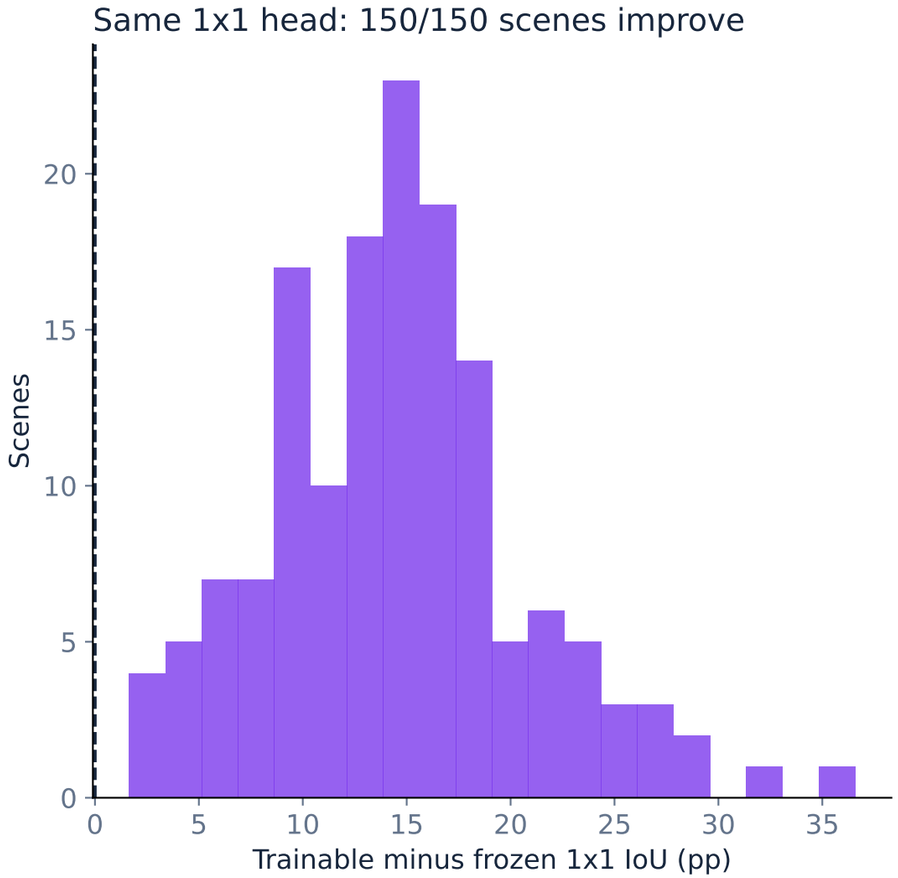


The end-to-end model misses 60.1% of valid GT vehicle cells.

Does the improvement hold across validation scenes?

Per-scene IoU pools valid-cell counts within each scene; distributions cover the same 150 scenes

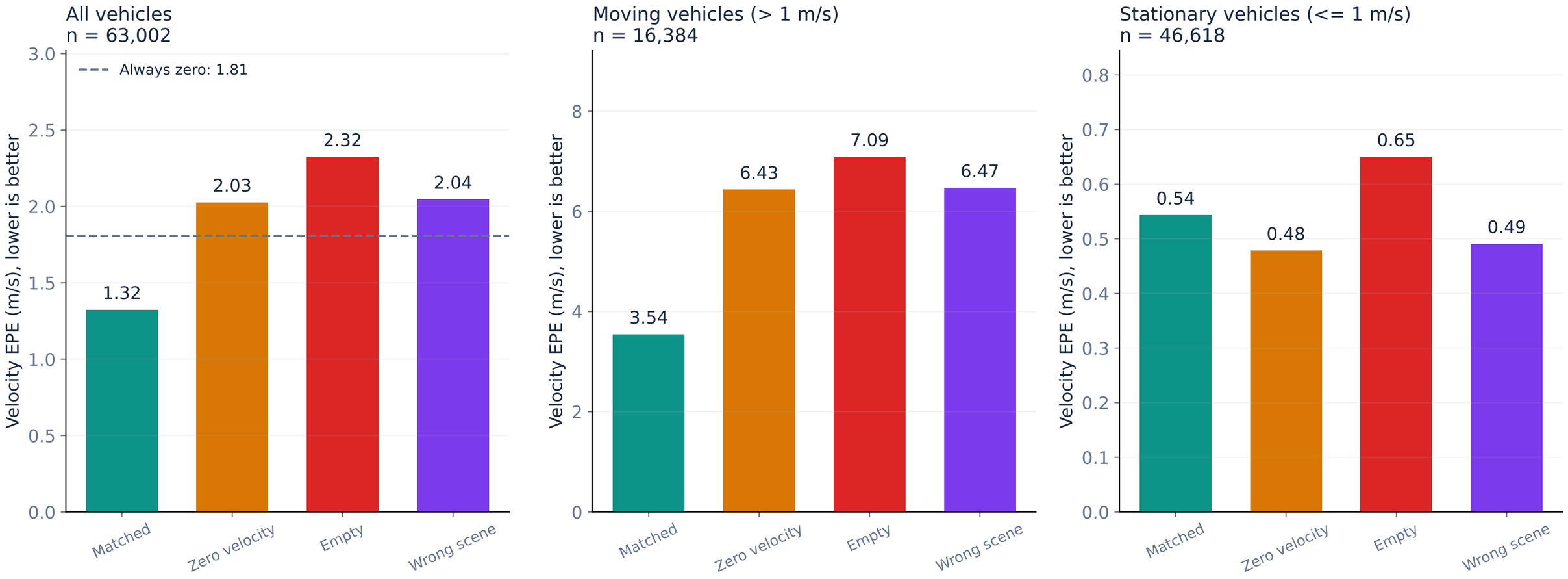




Scene results describe one trained seed; they do not measure training-seed uncertainty.

Radar contributes to vehicle velocity estimation

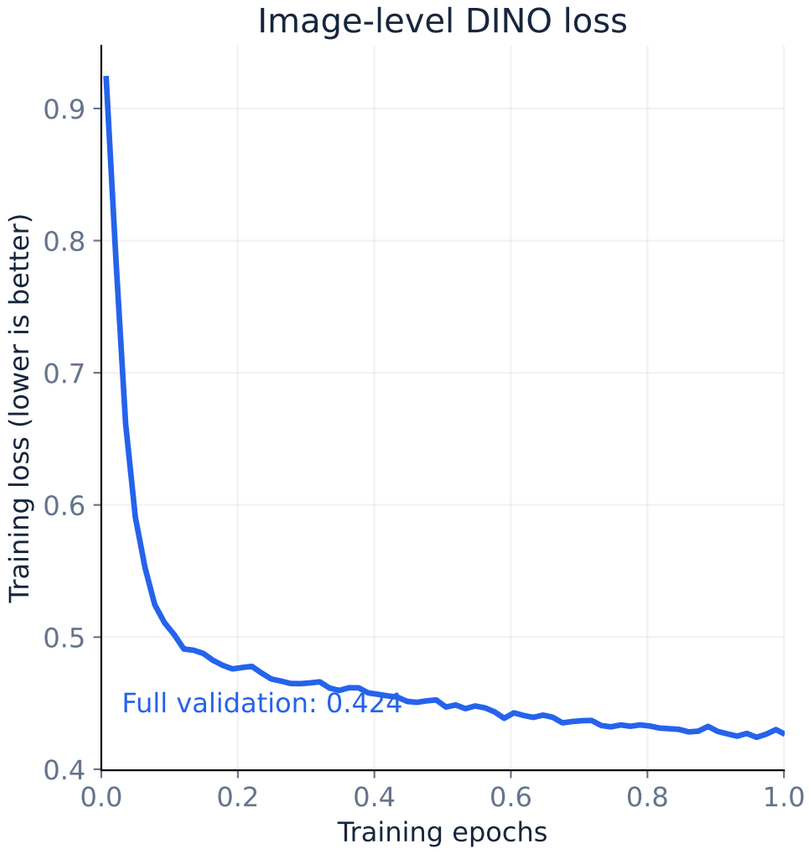
Frozen BEV features + trained linear velocity readout at GT box locations; all 6,019 validation frames

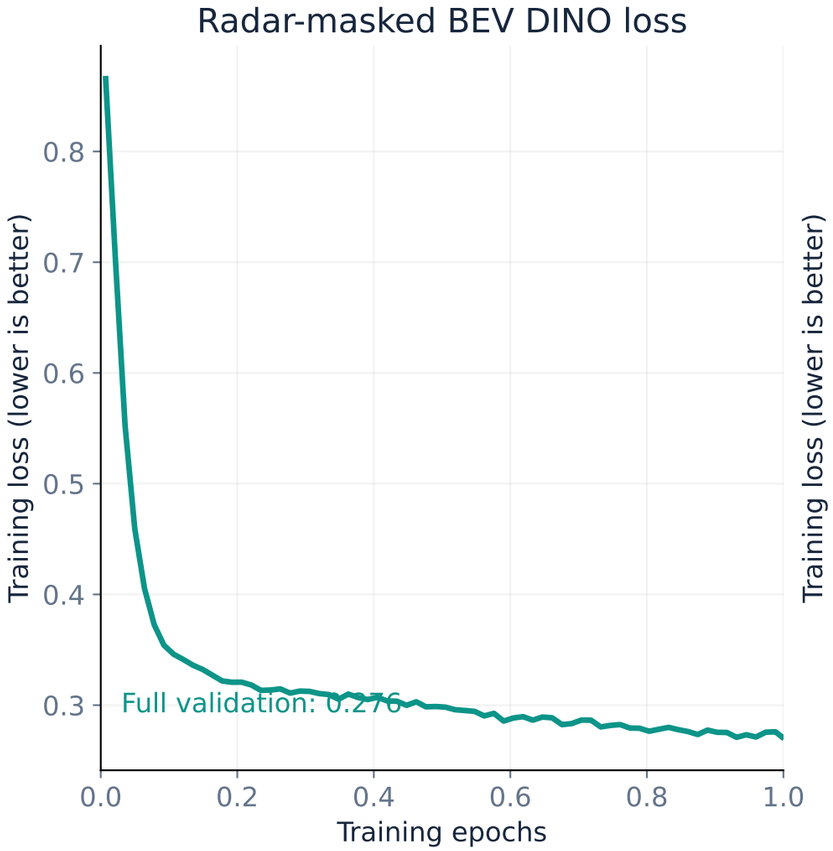


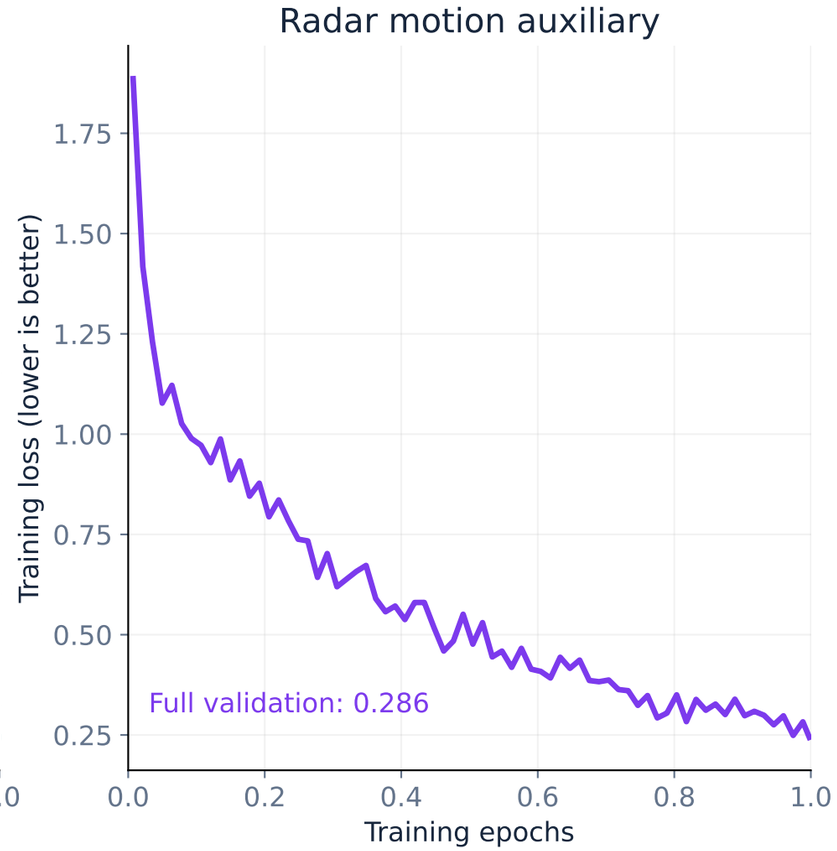
Aligned radar helps most for moving vehicles; stationary errors do not improve uniformly.

Pretraining learns the requested teacher objectives

Five sweeps, no depth: full training epoch; curves use non-overlapping 100-update means



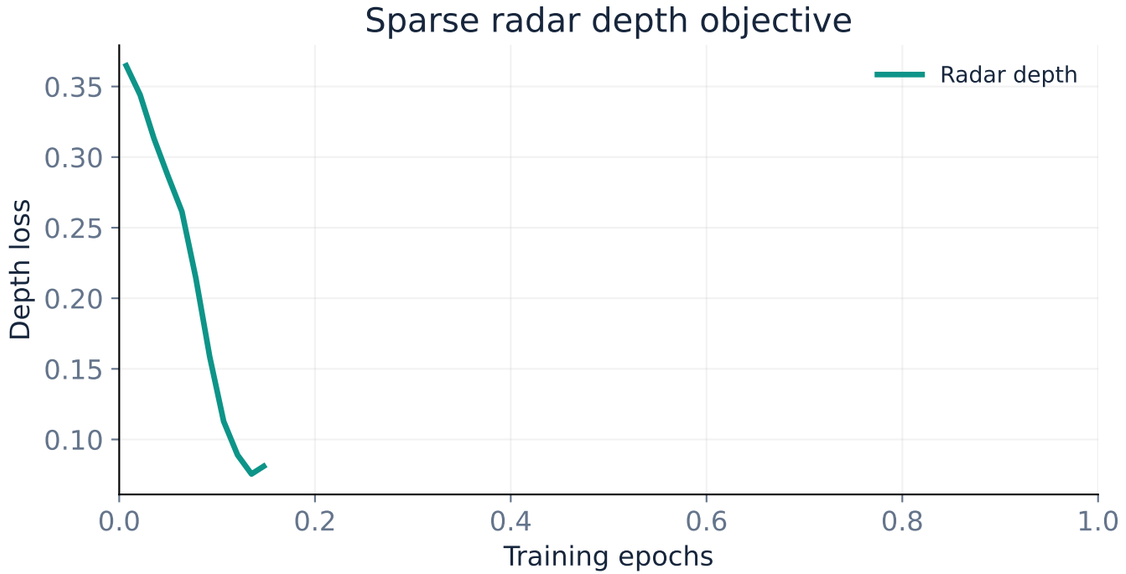


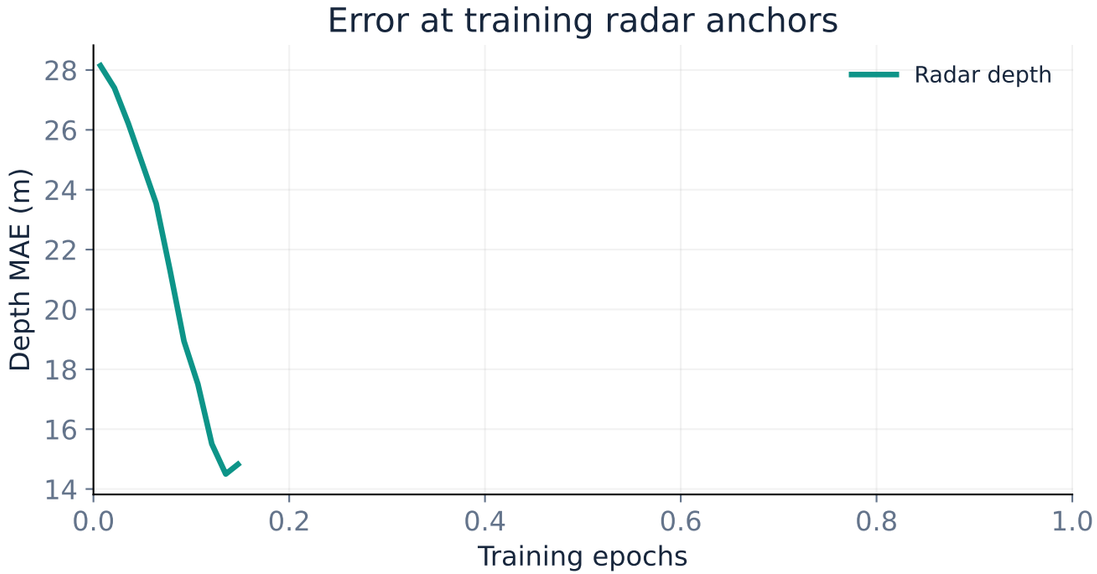


Teacher-feature matching and radar reconstruction improve; downstream GT evaluation remains necessary.

Depth experiment: current evidence and the next test

Live-run snapshot at 2026-10-16 12:09 UTC; displayed curves are training diagnostics at sparse radar anchors





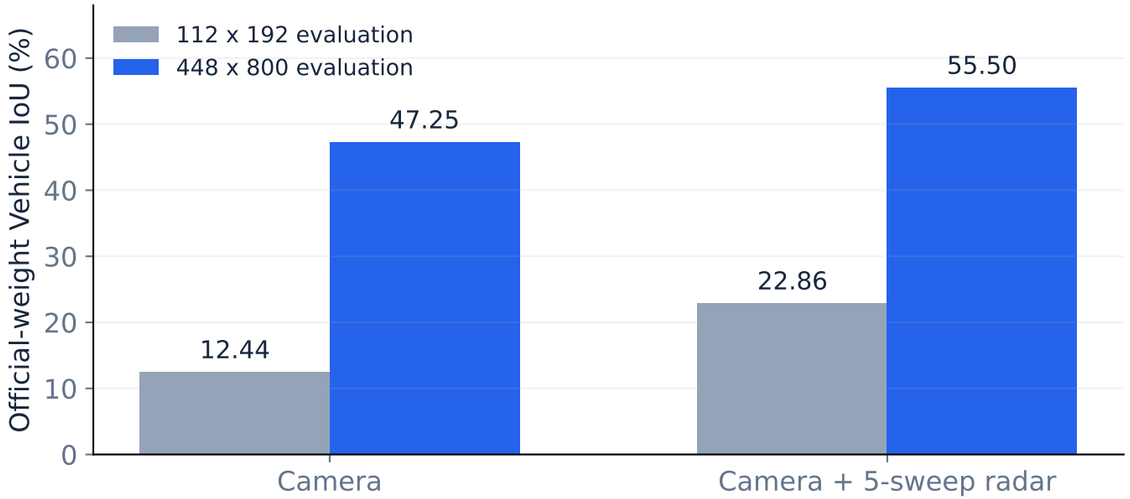
Radar depth: running; 15.3% training exposure
Radar + Depth Anything V2: no training records available at this snapshot

Treatment	Frozen 1x1 Vehicle IoU	End-to-end Vehicle IoU
No depth	10.45%	30.02%
Radar depth	Pending	Pending
Radar + Depth Anything V2	Pending	Pending

Depth training also changes image-to-BEV lifting; sparse-anchor errors are not independent dense-depth accuracy.

Next: a fair comparison at the standard resolution

Current findings motivate controlled tests; they do not yet establish the root cause of the IoU gap



Same official weights, different inference resolutions.
This tests resolution sensitivity; it does not isolate our training failure.

Corrected references from local summaries: 47.25% camera / 55.50% camera + radar.

Formal experiment

448 x 800 cameras + ImageNet-pretrained ResNet101

Trainable camera encoder + both DINO distillation terms

10-sweep VoxelNet radar input; measured batch / accumulation

Matched downstream budget, splits, augmentation and seeds

Frozen 1x1 probe + end-to-end Vehicle IoU

Same architecture without DINO as a pretraining control

Finish the depth screen, then test pretrained initialization and resolution under a matched protocol.